Automatic Power Factor Correction & Harmonic Mitigation System

PROJECT REPORT

submitted by

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in partial fulfilment of the requirements for the course

23ELC302 – Embedded System Design using ARM

of

BACHELOR OF TECHNOLOGY

IN

ELECTRICAL AND COMPUTER ENGINEERING



AMRITA SCHOOL OF ENGINEERING, COIMBATORE

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COIMBATORE – 641 112

November 2026

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Acknowledgements

We are grateful to our supervisor, Dr. Anand R, Assistant Professor (Sr. Gr.), Department of Electrical and Electronics Engineering, for guidance throughout this project, especially on the embedded design of the controller and the interpretation of the power quality measurements.

We thank Dr. Balamurugan S., Chairperson, Department of Electrical and Electronics Engineering, for the laboratory facilities that made it possible to build and test the hardware.

We also thank the faculty members and technical staff of the department for their help during the bench tests, and our families and friends for their constant encouragement.

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Abbreviations

ADC	Analog-to-Digital Converter
AHB	Advanced High-performance Bus
AHF	Active Harmonic Filter
APB	Advanced Peripheral Bus
APFC	Automatic Power Factor Correction
ARR	Auto-Reload Register (timer period minus one)
CCR	Capture/Compare Register (sets the PWM duty)
CMSIS	Cortex Microcontroller Software Interface Standard
DF2T	Direct Form II Transposed (IIR filter structure)
DFT	Discrete Fourier Transform
DMA	Direct Memory Access
DPF	Displacement Power Factor
FPU	Floating-Point Unit
GPIO	General-Purpose Input/Output
HAL	Hardware Abstraction Layer
HSI	High-Speed Internal oscillator
HT/TC	Half-Transfer / Transfer-Complete (DMA events)
ISR	Interrupt Service Routine
MCU	Microcontroller Unit
NVIC	Nested Vectored Interrupt Controller
PCC	Point of Common Coupling
PF	Power Factor
PLL	Phase-Locked Loop
PWM	Pulse-Width Modulation
Q15	Fixed-point format: value $\times 2^{15}$ stored in 16 bits
RMS	Root Mean Square
SSR	Solid-State Relay
THD	Total Harmonic Distortion

TRGO	Trigger Output (timer event routed to another peripheral)
UART	Universal Asynchronous Receiver-Transmitter
VOS	Voltage Output Scaling of the internal regulator
WFI	Wait For Interrupt (Cortex-M instruction)

Abstract

Many everyday single-phase loads, such as laptop adapters and phone chargers, lower the power factor in two different ways. Some draw a fundamental current that is shifted in phase from the voltage, while rectifier front ends draw pulsed currents rich in harmonics. A capacitor bank can supply the first kind of demand but cannot remove the second. This report describes a two-stage power quality system built around an STM32F401RE microcontroller on a NUCLEO-F401RE board. It addresses both problems with one embedded controller.

Stage 1 is an automatic power factor correction (APFC) unit. It switches capacitors of 2.5, 10 and 20 μF through two relay modules and one solid-state relay. Stage 2 is a demonstration active harmonic filter (AHF): a TB6612FNG H-bridge driven by 80 kHz PWM produces a current shaped to cancel harmonics 2 to 15 of the load current. In this prototype the AHF is not connected to the mains. The bridge drives a separate low-voltage test circuit, made of a 10 mH inductor, a second current sensor and a 10 W resistor, which shows that the required opposite harmonic current can be generated and tracked. A ZMPT101B module senses the voltage, and two ACS712-05B Hall-effect modules sense the load current and the AHF output current.

The firmware is bare-metal and written at register level with CMSIS, with no HAL and no operating system. A timer triggers the ADC at 200 samples per grid cycle, and DMA moves the samples into a double buffer. A software phase-locked loop keeps the sampling coherent with the mains. Every 100 ms the firmware folds five cycles into one, runs a table-based DFT and computes the rms values, active, reactive and distortion power, power factor, displacement power factor and THD. In the same window it runs a per-harmonic closed loop with online plant identification for the AHF. Every 0.5 s it chooses a capacitor combination and sends a telemetry frame to a Python dashboard over USB/UART.

In a small number of bench tests, the current THD before compensation ranged from 15.9% to 53.5%, depending on the load. The predicted current THD after compensation was between 3.2% and 14.4%. These after values are not measured. They are computed as the per-harmonic sum of the measured load current and the measured AHF output current, which is the grid current that would flow if the AHF current were injected at the connection point. In the dashboard snapshots, the measured AHF current followed its target waveform with a correlation of 0.976–0.982. The capacitor controller correctly held off switching when a load's reactive demand was below half of the smallest capacitor step.

CHAPTER 1

Introduction

1.1 Power Quality of Single-Phase Loads: Challenges and Proposed Solution

A modern household or laboratory supply feeds many loads that are not simple resistors. Motors and fans draw a current that lags the voltage. Switched-mode power supplies, such as laptop adapters and phone chargers, rectify the mains onto a capacitor and draw current only in short intervals near the voltage peaks. Both effects make the apparent power larger than the active power, so more current flows in the wiring than the useful power alone would need. The classical remedy, a capacitor bank, corrects only the first effect.

This project builds a two-stage power quality system around one STM32F401RE microcontroller. The microcontroller measures the voltage and current coherently with the grid and separates the two causes of a low power factor. It then acts on each one: a switched capacitor bank supplies the reactive power, and an H-bridge driven harmonic by harmonic generates the opposite harmonic current that would cancel the load's distortion. A Python dashboard shows the results and accepts operator commands. The system is a single-phase bench prototype. The harmonic stage uses a 12 V motor-driver H-bridge that drives its own test circuit and is not connected to the mains, so its effect on the grid current is computed from measurements rather than observed.

1.2 Background

It helps to separate two causes of a low power factor [1, 2]:

- **Displacement.** The fundamental (50 Hz) current is shifted in phase from the voltage. This requires fundamental reactive power Q_1. Shunt capacitors are the classical remedy, because they supply leading reactive power locally.
- **Distortion.** The current contains harmonics, which are components at integer multiples of 50 Hz. They add a distortion power D that is neither active nor reactive. A capacitor cannot cancel it. An active filter can, by injecting a current that carries the opposite harmonics [3, 4].

The true power factor PF $= P/S$ includes both effects. The displacement power factor DPF $= \cos\varphi_1$ includes only the first. A system that has to raise the true power factor must therefore measure and treat the two separately.

The STM32F401RE is an Arm Cortex-M4F microcontroller with a single-precision FPU, 512 KB of Flash and 96 KB of SRAM, running at up to 84 MHz [5]. Its timers, ADC and DMA controller allow the mains to be sampled at a precise rate without CPU involvement, and its FPU makes per-window DFTs and control loops practical in floating point.

1.3 Problem Statement

Capacitor-based correction alone cannot improve the power factor of a load whose penalty comes mostly from harmonics, and switching capacitors without care can damage relay contacts. The problem is to build a single embedded controller that:

(a) measures voltage, current, power, power factor and harmonic content accurately enough to tell displacement from distortion;

(b) decides when switching a capacitor step actually improves the displacement factor, and switches it in a way that limits contact stress; and

(c) drives an active harmonic filter in closed loop, so that its output current follows the harmonic content that must be cancelled.

All of this must run in real time on a low-cost microcontroller.

1.4 Motivation

The two correction methods complement each other. Reactive power at 50 Hz is usually large and steady and can be supplied passively by capacitors at almost no loss. Harmonic currents are smaller but vary with the load. Handling them with an active stage, while capacitors handle the reactive power, keeps the current rating of the active stage small. A combined design is also a good way to learn the embedded techniques the problem needs: hardware-timed sampling, DMA, interrupt priorities, fixed-point PWM, digital signal processing and closed-loop control. All of them must work together within a strict timing budget.

1.5 Research Questions

The work addresses the following questions:

1. How can a single low-cost microcontroller sample voltage and current coherently with the grid, accurately enough to separate displacement from distortion?
2. When does switching a capacitor step actually improve the power factor of a given load, and when should the controller hold instead?
3. Can a low-power H-bridge, controlled harmonic by harmonic, generate the opposite harmonic current needed to cancel a load's distortion?
4. How should the real-time work be divided between hardware peripherals, interrupts and the main loop so that sampling, control and telemetry all meet their deadlines?
5. How far could the current THD be reduced if the generated current were injected, and what limits that prediction?

1.6 Objectives

1. Acquire one voltage and two currents synchronously at 200 samples per grid cycle, using a timer-triggered ADC with DMA double buffering on the STM32F401RE.
2. Lock the sampling clock to the grid frequency with a software phase-locked loop, so that every analysis window holds an integer number of cycles.
3. Compute the rms values, P, Q_1, S, D, PF, DPF, THD and harmonics 1–15 on the microcontroller every 100 ms.
4. Implement an APFC controller that selects the best combination of 2.5, 10 and 20 μF capacitors, with hysteresis, minimum on/off times and point-on-wave relay timing.
5. Implement a selective-harmonic AHF controller for harmonics 2–15, driving a TB6612FNG H-bridge with 80 kHz sign-magnitude PWM.
6. Stream all results to a Python dashboard over USB/UART and accept manual capacitor and AHF commands from it.

CHAPTER 2

Literature Survey

2.1 Literature Review of Related Works

When currents are distorted, the familiar triangle $S^2 = P^2 + Q^2$ no longer accounts for all of the apparent power. IEEE Std 1459 sets out definitions for single-phase and three-phase circuits under sinusoidal and non-sinusoidal conditions [1]. Among them, it separates the fundamental active and reactive powers from the non-fundamental components. In this project, the quantity that the capacitors act on is the fundamental reactive power Q_1. The remainder, the distortion power $D = \sqrt{S^2 - P^2 - Q_1^2}$, is the part only the active filter can reduce. Dugan *et al.* [2] explain how harmonic currents arise in rectifier loads and why the true power factor of such loads stays low even when the displacement factor is close to one.

IEEE Std 519-2022 [6] sets design goals for harmonic voltage and current at the point of common coupling between a utility and its customers. IEC 61000-3-2 [7] limits the harmonic current emissions of individual equipment with an input current of up to 16 A per phase. IEC 61000-4-7 [8] defines how harmonic measuring instruments should work, including synchronisation of the measurement window to the supply frequency. The prototype in this report does not claim compliance with any of these standards. They are cited because they define the quantities it reports and the measurement practice it approximates, such as grid-synchronised windows of whole cycles.

A shunt capacitor C on a voltage V at frequency f supplies a reactive power $Q_C = 2\pi f C V^2$. Stepped capacitor banks that are switched automatically are a common way to hold the displacement factor near a target [2]. The literature also stresses the transients caused by capacitor switching. Energising a capacitor whose voltage differs from the instantaneous line voltage produces a large inrush current, and switching back-to-back makes this worse [2]. This is the practical motivation for the minimum on/off times and the point-on-wave switching used in the present controller.

Akagi, Kanazawa and Nabae introduced instantaneous reactive power theory for three-phase compensators built from switching devices [9]. This work became a foundation for active filter control. Later reviews by Akagi [3] and by Singh, Al-Haddad and Chandra [4] classify active filters by topology (shunt, series, hybrid), by supply type and by control strategy. They show that shunt active filters inject compensating

currents in parallel with the load. Mohan, Undeland and Robbins [10] cover the converter fundamentals involved: PWM bridges, the averaged output voltage, and the role of the series inductance that turns the bridge voltage into a controlled current.

A filter that reproduces the whole distortion waveform needs a current controller with high bandwidth. Mattavelli proposed an alternative: a closed loop for each selected harmonic, using line-current measurements and controllers in synchronous or equivalent stationary frames [11]. The paper reports full compensation of the selected harmonics even with a significant delay in the current control loop. The AHF in this project follows the same principle. Each harmonic from 2 to 15 has its own phasor loop. The controller also measures the complex gain of the bridge and its output circuit at each harmonic online, so that a delay or the attenuation of the inductor is corrected harmonic by harmonic.

Grid-connected converters need the phase and frequency of the supply. Teodorescu, Liserre and Rodríguez [12] review synchronisation methods for single-phase and three-phase converters. These include Fourier-based methods, phase-locked loops and frequency-locked loops. The present system uses a simple Fourier-based software PLL. The phase of the voltage fundamental, obtained from the DFT, is driven to zero by adjusting the sampling timer, so that sampling stays coherent with the grid.

The bilinear transform with frequency pre-warping, the direct-form filter structures, and the leakage-free behaviour of the DFT when a window holds whole cycles are standard results of discrete-time signal processing [13]. On the hardware side, the STM32F401 reference manual [14] documents the timer trigger outputs, the ADC scan mode with external triggering, the circular DMA with half-transfer interrupts and the NVIC priorities that the firmware relies on.

2.1.1 Summary of the Literature Review

Table 2.1 summarises how each source is used in this project.

The literature treats capacitor banks and active filters mainly as separate, high-power equipment controlled by dedicated hardware. Less attention goes to showing, on one low-cost microcontroller, the complete chain from coherent sampling to the decision on displacement versus distortion and the two corresponding actuators. This project fills that gap for a single-phase prototype. Its contribution is the embedded integration: hardware-timed sampling locked to the grid, measurement on the MCU, capacitor switching aware of hold reasons, and a selective-harmonic AHF loop with online plant identification. It does not claim a new filter topology or grid-rated performance.

Table 2.1: Summary of the most relevant works

Source	**Main content**	**Use in this project**
IEEE Std 1459 [1]	Power definitions under non-sinusoidal conditions	Definitions of P, Q_1, S, D, PF and DPF
Dugan *et al.* [2]	Harmonics, capacitor application, switching transients	Motivation for the two stages and for relay protection
IEEE Std 519-2022 [6]	Harmonic limits at the PCC	Context for the voltage-THD indicator only; no compliance claim
IEC 61000-3-2 / 61000-4-7 [7, 8]	Equipment emission limits; harmonic measurement practice	Context for coherent, synchronised windows
Akagi *et al.* [9], Akagi [3], Singh *et al.* [4]	Active filter theory and topologies	Choice of a shunt AHF
Mattavelli [11]	Closed-loop selective harmonic compensation	Per-harmonic phasor loop
Teodorescu *et al.* [12]	Grid synchronisation methods	Fourier-based software PLL
Oppenheim and Schafer [13]	DFT, IIR design, bilinear transform	DFT, Butterworth filters, DF2T
Mohan *et al.* [10]	PWM converters	H-bridge drive and series inductor

CHAPTER 3

Methodology

3.1 Overview

The system is built in three layers. The hardware layer senses the mains voltage and two currents, switches the capacitor steps and drives the H-bridge. The firmware layer on the STM32F401RE samples the signals in step with the grid, computes the power quality quantities every 100 ms, and runs both control loops. The PC layer is a dashboard that displays the results and sends operator commands. This chapter describes the overall architecture, the hardware, the firmware, the signal processing, the two control stages, the communication link and the dashboard.

3.2 Overall System Architecture

Fig. 3.1 gives the circuit diagram of the prototype. Three analog signals enter the microcontroller: the mains voltage, the current drawn from the grid, and the output current of the AHF. The firmware drives three outputs: the GPIO lines that switch the capacitor steps, two PWM channels for the H-bridge, and a serial link to the PC. Measurement, control and protection all run on the STM32F401RE. The PC only displays the results and sends operator commands; if it is disconnected, the board keeps correcting. The main points of the circuit are:

- **Sensing.** ACS712-05B #1 is in series with the line conductor and ZMPT101B is across L–N. Both are supplied from the 5 V rail of the NUCLEO board. The ACS712 output reaches PA0 through the 10 kΩ/20 kΩ divider, and the ZMPT101B output goes to PA1.
- **Load and capacitor bank.** Both connect at the point of common coupling (PCC) downstream of sensor I. Sensor I therefore measures the load current plus the current of any capacitor that is switched in. This is why the APFC algorithm adds the switched capacitance back when it estimates the load's own reactive demand.
- **Capacitor switching.** K1 (2.5 μF, PB0) and K2 (10 μF, PB1) are 5 V active-LOW relay modules rated 10 A. K3 (20 μF, PB2) is a zero-cross, active-HIGH SSR. The capacitors are rated 400 VAC.

- **AHF output.** The TB6612FNG takes its logic supply (VCC) of 3.3 V from the NUCLEO board and its motor supply (VM) of 12 V from a separate DC supply. STBY and PWMA are tied HIGH, and AIN1/AIN2 are driven from PB6/PB8. Output AO1 feeds the 10 mH inductor, ACS712-05B #2 (supplied at 5 V, output to PA4) and the 10 W resistor that form the AHF test circuit. This circuit is not connected to the mains (Section 3.3.4).

- **Communication.** USART2 (PA2/PA3) connects to the PC dashboard over USB at 115 200 bit/s.

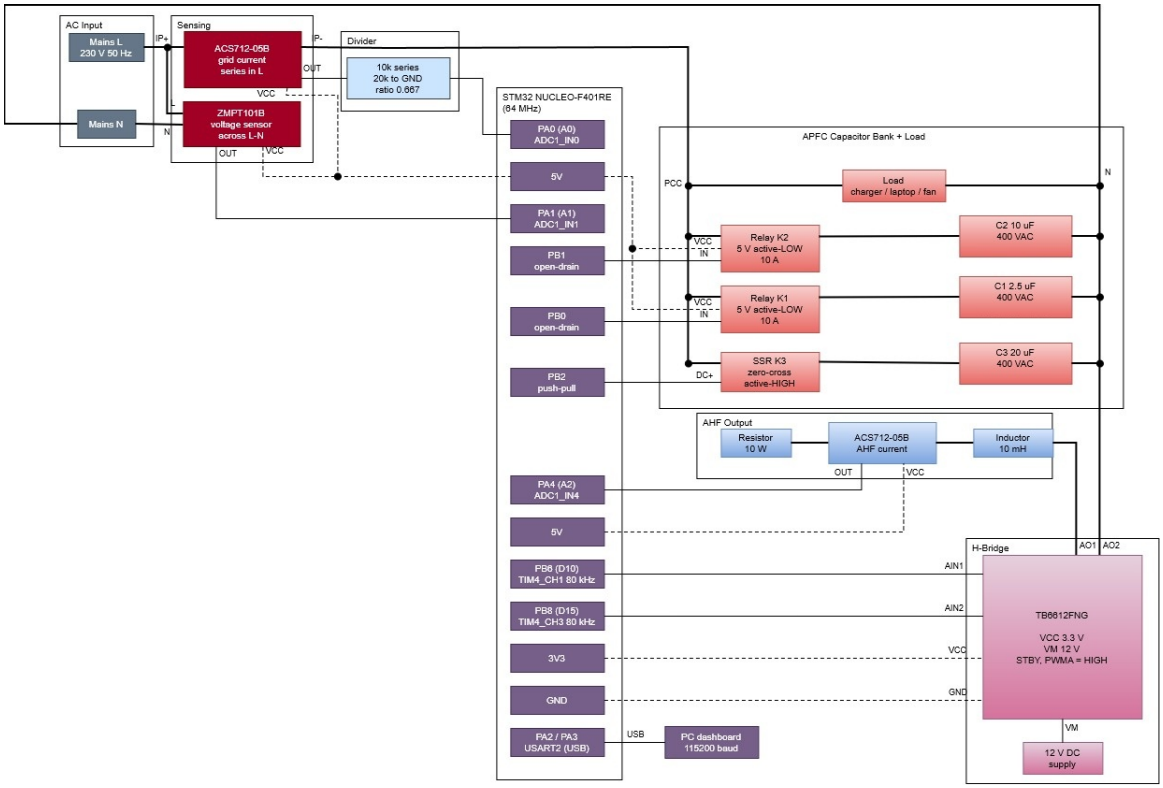


Figure 3.1: Circuit diagram of the prototype: sensing, APFC capacitor bank with relays and SSR, AHF output stage and NUCLEO-F401RE connections

3.3 Hardware Design

3.3.1 Components and Pin Assignment

Table 3.1 gives the pin configuration as set in the firmware. The two relay modules have active-LOW inputs referenced to 5 V. If they were driven push-pull from a 3.3 V pin, the HIGH level would leave about 1.7 V across the opto-coupler LED of the module, which may keep the relay from releasing fully. The firmware therefore drives PB0 and PB1 open-drain, with the internal pull-up disabled. The pin either sinks

current (relay ON) or floats, so that the module's own pull-up takes the input to 5 V (relay OFF). This requires 5 V-tolerant pins. The SSR on PB2 is active-HIGH and driven push-pull. At start-up the firmware writes the OFF level before it switches the relay pins to output mode, so the relays cannot click on during a reset.

Table 3.1: Pin assignment of the STM32F401RE (NUCLEO-F401RE)

Pin (header)	Mode	Function
PA0 (A0)	Analog	ADC1_IN0, grid current (sensor I, via divider)
PA1 (A1)	Analog	ADC1_IN1, voltage (ZMPT101B)
PA4 (A2)	Analog	ADC1_IN4, AHF current (sensor A, direct)
PA2 / PA3	AF7	USART2 TX / RX to the ST-LINK USB bridge
PA5	Output	LED LD2 heartbeat, toggled every 0.5 s
PB0, PB1	Output, open-drain	Relay modules for 2.5 μF and 10 μF (5 V active-LOW)
PB2	Output, push-pull	SSR for 20 μF (active-HIGH)
PB6 (D10)	AF2	TIM4 CH1 to a TB6612FNG input
PB8 (D15)	AF2	TIM4 CH3 to a TB6612FNG input

3.3.2 Sensing and Signal Conditioning

Voltage. The ZMPT101B module contains a small voltage transformer and an op-amp stage. It outputs a low-voltage copy of the mains, centred on a DC offset, and is supplied from the 5 V rail of the NUCLEO board. The module is connected to PA1 (ADC1 channel 1). Its ratio was calibrated as 1014 mains volts per ADC volt (`ZMPT_V_PER_V`), so a 230 V supply (325 V peak) appears as about ±398 ADC counts around the offset.

Current. Both currents are measured with the 5 A version of the ACS712 Hall-effect sensor, which has a sensitivity of 185 mV/A [15]. Its output rests at half its 5 V supply (2.5 V) at zero current, and it is ratiometric. Sensor I, on PA0, measures the current drawn from the grid. Sensor A, on PA4, measures the current that the AHF drives through its test circuit.

The STM32 ADC accepts 0–3.3 V. With the zero point at 2.5 V, a direct connection leaves only 0.8 V of headroom for positive current, which is about 4.3 A. In an earlier test a 6 A load clipped on every positive half-cycle and produced a false THD reading of 22.9%. Sensor I was therefore given a resistive divider of 10 kΩ in series and 20 kΩ to ground. This gives a ratio of 0.6667, a zero point near 2068 counts, and a measurable range of about −13.5 A to +13.2 A. Sensor A is connected directly, because the AHF current is small.

Scaling. The 12-bit ADC gives 0.806 mV per count. Table 3.2 lists the conversion

factors set in the firmware. The firmware does not assume fixed zero offsets. It subtracts the mean of each window, which equals the DC offset exactly when the window holds whole cycles. This cancels resistor tolerance and supply drift at the zero point.

Table 3.2: Channel scaling constants

Channel	Pin	Formula	Value per count
K_I (grid current)	PA0	$\frac{3.3/4095}{0.185 \times 0.6667}$	6.534 mA
K_A (AHF current)	PA4	$\frac{3.3/4095}{0.185 \times 1}$	4.356 mA
K_V (voltage)	PA1	$\frac{3.3}{4095} \times 1014$	0.817 V

Noise. With no current or voltage present, the readings still fluctuate. According to the study guide, the noise measured on this hardware was about 31 mA rms on sensor I and about 6 V rms on the voltage channel with the mains disconnected [16]. Section 3.5 describes how the firmware separates this noise from the real signal.

3.3.3 Capacitor Bank

Table 3.3 gives the size and switching device of each capacitor step. The reactive power per microfarad at 230 V and 50 Hz is $k = 2\pi f V^2 \times 10^{-6} = 16.62\,\text{var}/\mu\text{F}$.

Table 3.3: Capacitor steps and switching devices

Step	Pin / device	Q_C at 230 V, 50 Hz	Current
2.5 μF	PB0, relay module (active-LOW)	41.5 var	0.18 A
10 μF	PB1, relay module (active-LOW)	166 var	0.72 A
20 μF	PB2, SSR (active-HIGH)	332 var	1.45 A

3.3.4 AHF Output Stage

In this prototype the H-bridge output is not connected to the mains. The bridge drives a series path made of a 10 mH inductor, the ACS712-05B sensor A and a 10 W resistor. The resistor loads the output so that the generated current can be sensed; its resistance was not recorded. From its 12 V supply, the bridge drives the required opposite harmonic current through this path, and sensor A measures it. This low-power arrangement closes the control loop and checks that the commanded current is actually produced, without connecting a motor-driver stage to the grid.

The firmware is configured to match (`AHF_SENSE_GRID = 0`). Sensor I measures the load current, which the AHF does not alter, and the current after compensation is formed in software as the per-harmonic sum $I_h + A_h$, with an injection ratio of 1. That sum is a *prediction* of the grid current that would flow if the measured AHF current were injected where the load connects. It is not a measurement of any current that flows in the prototype.

TIM4 runs at 64 MHz with ARR = 799, giving 80 kHz, below the 100 kHz input limit of the TB6612FNG [17]. That provides 800 duty steps (0.125% each). For a positive duty, one bridge input is held permanently HIGH by setting its compare value to `0xFFFF`, larger than ARR. The other input is pulsed LOW for d of the p ticks. The bridge then alternates between drive ($+V_M$) and short brake (0 V), so the average output voltage is $V_M\, d/p$. For a negative duty the two inputs swap roles. In STANDBY both compare values are 0, so both inputs are LOW and the bridge outputs are off. The firmware never drives the PWMA and STBY pins of the TB6612FNG; they are tied HIGH in hardware (Fig. 3.1), so the bridge is controlled entirely through AIN1 and AIN2.

3.4 STM32 Firmware

3.4.1 On-Chip Resources

The STM32F401RE has an Arm Cortex-M4 core with a single-precision FPU, 512 KB of Flash and 96 KB of SRAM, and runs at up to 84 MHz [5]. The hardware FPU is what makes it practical to run DFTs, IIR filters and complex-number control loops in floating point every 100 ms. The firmware enables it before the first floating-point instruction.

The firmware uses no HAL. Every peripheral is configured by writing its registers directly, using the CMSIS device header `stm32f401xe.h` [14]. This was chosen for three reasons. It gives full visibility and control of the timer, ADC and DMA synchronisation. It keeps the code small with deterministic interrupt timing. And it makes every configuration choice explicit and open to review. Table 3.4 lists the on-chip resources used.

3.4.2 Clock Configuration

The firmware runs from the 16 MHz internal HSI oscillator through the main PLL, with $M = 16$, $N = 256$ and $P = 4$:

$$f_{\mathrm{SYSCLK}} = \frac{16\,\mathrm{MHz}}{16} \times \frac{256}{4} = 64\,\mathrm{MHz}. \tag{3.1}$$

Table 3.4: On-chip resources used by the firmware

Peripheral	Bus (clock)	Purpose
GPIOA, GPIOB	AHB1 (64 MHz)	Analog inputs, UART pins, LED, relay outputs, PWM pins
DMA2 Stream 0, Channel 0	AHB1	ADC1 results to RAM
DMA1 Stream 6, Channel 4	AHB1	USART2 transmission (telemetry)
TIM2 (32-bit)	APB1 timers (64 MHz)	Sample clock, grid lock, 10 kHz interrupt
TIM4 (16-bit)	APB1 timers (64 MHz)	80 kHz PWM for the H-bridge
USART2	APB1 (32 MHz)	115 200 bit/s link via the ST-LINK virtual COM port [18]
ADC1	APB2 (64 MHz; ADC 16 MHz)	Three-channel scan conversion
PWR, FLASH	–	Voltage scaling, Flash wait states
NVIC, SCB	Core	Interrupt priorities, FPU enable

APB1 runs at 32 MHz. Because its prescaler is not 1, the timers on APB1 receive twice that, 64 MHz. APB2 runs at 64 MHz and the ADC clock is APB2/4 = 16 MHz. Before switching to the PLL, the Flash is set to two wait states, with prefetch and caches enabled, and the regulator to voltage scale 2.

The clock is 64 MHz rather than the maximum 84 MHz because 64 MHz divides exactly into every frequency the design needs. There are 6400 timer ticks per 10 kHz sample, 800 ticks per 80 kHz PWM period, exactly eight PWM periods per sample, and a 16 MHz ADC clock.

3.4.3 Sampling Design

TIM2 runs with prescaler 0 and ARR = 6399, which gives a 10 kHz update rate: 200 samples per 50 Hz cycle. The highest harmonic processed is the 15th (750 Hz), well below the 5 kHz Nyquist frequency. When the number of samples per cycle is an integer and the analysis covers whole cycles, every harmonic falls exactly on a DFT bin, so no spectral leakage occurs and no window function is needed [13]. Because the real grid frequency drifts (49.02–49.45 Hz was observed on the project supply [16]), a software phase-locked loop (PLL) continuously adjusts the TIM2 period from the phase of the voltage fundamental, to keep exactly 200 samples per cycle. The PLL also holds the positive voltage peak at sample $k = 0$ of every cycle.

The update event of TIM2 is routed as its trigger output (TRGO, MMS = 010). This one signal is used twice:

1. ADC1 starts a scan on the rising edge of TIM2_TRGO (EXTSEL = 6). The

sample instants are therefore set by hardware, with no software jitter.

2. TIM4 is configured as a slave of TIM2 in reset mode (internal trigger ITR1), so every sample restarts the PWM counter. The eight PWM periods inside each sample stay aligned with the ADC sampling instant, and the PWM ripple cannot alias into the measured harmonics.

ADC1 scans IN0, IN1 and IN4 in that order, with the longest sampling time of 480 cycles. This long time lets the sample-and-hold capacitor settle through the source impedance of the sensor modules and the 6.7 kΩ divider. One conversion takes $(480 + 12)/16\,\text{MHz} = 30.75\,\mu\text{s}$ and a full scan takes $92.25\,\mu\text{s}$. That fits within the $100\,\mu\text{s}$ sample period, and also within the $95.2\,\mu\text{s}$ period at the PLL's upper limit of 52.5 Hz. A compile-time assertion in the firmware stops the build if this margin is ever violated.

Because the channels are converted one after another, the voltage is sampled $30.75\,\mu\text{s}$ after the current, which is a phase shift of 0.55° at 50 Hz. The firmware rotates the voltage phasor back by this angle before it computes P_1 and Q_1. For sensor A the skew needs no separate correction, because the online plant identification of the AHF absorbs it.

3.4.4 ADC–DMA Double Buffering

Reading 30 000 conversions per second with the CPU would cost 30 000 interrupts per second. Instead, DMA2 Stream 0, Channel 0 copies each 16-bit result from the ADC data register into a circular buffer of 6000 half-words. The values are interleaved as I, V, A, I, V, A, and so on. The buffer is used as a ping-pong pair (Fig. 3.2). The half-transfer interrupt signals that the first half (1000 scans) is ready, and the transfer-complete interrupt signals the second half. Each half is one *window* of 1000 samples per channel: 100 ms, or five grid cycles. While the DMA fills one half, the main loop copies the other half into a separate array (`snap`) and processes the copy.

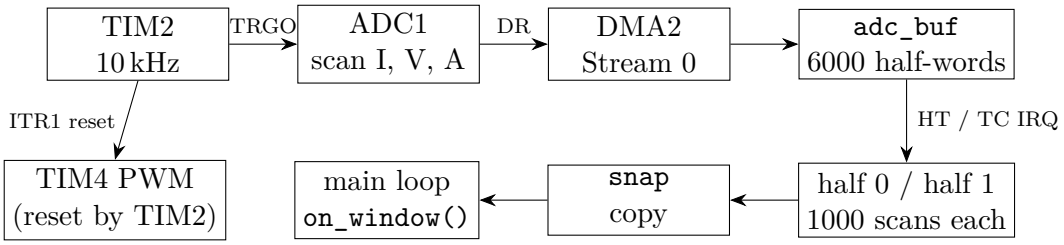


Figure 3.2: Hardware-timed acquisition chain and DMA ping-pong buffering

3.4.5 Interrupts, Priorities and Shared Data

Table 3.5 lists the interrupts and their NVIC priorities (a lower number is more urgent). TIM2 has the highest priority because it writes the next PWM duty value and times the relay coils to within one sample. It is kept short so that it never delays anything noticeably. The heavy processing runs in the main loop, so no ISR blocks the time-critical ones.

Table 3.5: Interrupt sources, rates and priorities

ISR	Rate	Priority	Work
`TIM2_IRQHandler`	10 kHz	0	Dither the period, step the AHF table, set PWM compares, timed relay coils
`DMA2_Stream0_IRQHandler`	10 Hz	1	Mark window ready, 100 ms tick, clean-window flag
`ADC_IRQHandler`	On error	1	Count overrun, request restart
`USART2_IRQHandler`	Per byte	2	Receive command characters
`DMA1_Stream6_IRQHandler`	2 Hz	3	Telemetry transmission finished

The firmware follows four rules for data shared between ISRs and the main loop:

- variables shared with an ISR are declared `volatile`;
- related variables, such as the lock masks, the PLL period, the pending relay masks and the window-ready flag, are read or changed inside short critical sections with interrupts disabled;
- a memory barrier (`__DMB()`) makes sure a newly computed PWM table is fully written before the flag that hands it to the ISR is set; and
- the AHF table is double-buffered. The ISR plays one copy while the main loop writes the other, and the swap happens only at sample 0 of a cycle.

When no window is waiting, the main loop executes `WFI`. The core then sleeps until the next interrupt, at most 100 μs later.

3.4.6 Initialisation and Main Loop

The firmware is written in C and organised in layers: constants with compile-time assertions; a DSP core (complex-number helpers, cosine/sine tables, Butterworth design, zero-phase filtering, DFT and THD, the AHF control step and the PLL update); hardware initialisation; the ISRs of Table 3.5; processing, control and telemetry functions; and `main()`.

After reset, `main()` performs the following steps in a fixed order:

1. enables the FPU;
2. `clock_init`: voltage scale 2, two Flash wait states, HSI, PLL start, switch to 64 MHz;
3. `gpio_init`: relays forced OFF first, then all pins configured;
4. `tables_init`: cosine/sine tables, the $\tan\varphi$ target, the AHF model, filter coefficients and responses;
5. `uart_init`, `tim2_init`, `adc_dma_init`, `tim4_init`; and
6. `nvic_init`, called last, so that no interrupt can fire before its peripheral is configured.

`sampling_sync_start` then arms the DMA, enables the ADC and starts TIM4 and TIM2. From that point on, samples flow without CPU involvement.

The main loop is cooperative and there is no operating system (Listing 3.1). On every pass it restarts the acquisition chain if an ISR reported an ADC overrun or DMA error, and services the relays. It then takes the ready-window index atomically. If a window is ready, it copies that half of the buffer and processes it; otherwise it sleeps with `WFI`.

Listing 3.1: Main loop of the firmware

```
for (;;) {
    if (resync_req) {
        resync_req = false;
        sampling_sync_start();
    }

    pfc_cmd_pending = false;
    relay_service(ms_ticks); /* applies manual locks immediately */

    __disable_irq();
    const int32_t half = win_ready;
    win_ready = -1;
    __enable_irq();

    if (half >= 0) {
        const bool clean = win_clean[half] != 0u;
        memcpy(snap, (const void *)&adc_buf[(uint32_t)half * N_WIN * N_CH],
            sizeof snap);
        on_window(clean);
    } else {
        __WFI();
    }
```

```
}
```

Each window is handled by `on_window()` (Listing 3.2). It runs the measurements, updates and applies the grid lock, and performs one AHF control step. Every fifth window (0.5 s) it averages the measurements and runs the APFC decision, provided at least three windows were averaged. It then sends a telemetry frame and toggles the LED. Two windows after every relay change are excluded from the average while the transient settles.

Listing 3.2: Window handler of the firmware

```
static void on_window(bool clean)
{
    process_window(snap, &mw);
    pll_update(av[1], bv[1], mw.vrms > 0.0f);
    grid_v_present = mw.vrms > 0.0f;
    pll_apply();
    ahf_step(clean);

    if (settle_win) settle_win--;
    else acc_add(&mw);

    if (++win_count >= TELEM_WIN) {
        win_count = 0;
        avg_t a;
        make_avg(&a, &mw);
        if (acc.n >= 3u) apfc_step(&a);
        telemetry_send(&mw, &a);
        acc_reset();
        GPIOA->ODR ^= GPIO_ODR_OD5;
    }
}
```

3.5 Signal Processing

3.5.1 Coherent Folding and DFT

Each window of five cycles is folded into one averaged cycle of 200 points. Sample k of the averaged cycle is the mean of samples k, $k+200$, $k+400$, $k+600$ and $k+800$. The repeating waveform of the load is unchanged, while random noise falls by a factor of $\sqrt{5} \approx 2.24$.

For the averaged cycle $x_0, \ldots, x_{199}$ and each harmonic $h = 1, \ldots, 15$:

$$a_h = \frac{2}{N} \sum_{k=0}^{N-1} x_k \cos \frac{2\pi h k}{N}, \qquad b_h = \frac{2}{N} \sum_{k=0}^{N-1} x_k \sin \frac{2\pi h k}{N}, \tag{3.2}$$

with $N = 200$. The rms value of harmonic h is $\sqrt{a_h^2 + b_h^2}/\sqrt{2}$. The cosine and sine values are precomputed in tables. For harmonic h, the table index advances by h per sample and wraps at 200, so no trigonometric function is called at run time. Only 15 harmonics are needed, so a direct DFT (15×200 multiply–adds per channel) is cheaper than an FFT. An FFT would compute 100 bins, and 200 is not a power of two. Each harmonic is held as a phasor $X = a - jb$, because $\mathrm{Re}\{Xe^{j\theta}\} = a\cos\theta + b\sin\theta$. A gain or phase shift then becomes a single complex multiplication.

3.5.2 Power Quantities

For voltage and current fundamentals with coefficients (a_v, b_v) and (a_i, b_i):

$$P_1 = \tfrac{1}{2}\left(a_v a_i + b_v b_i\right), \qquad Q_1 = \tfrac{1}{2}\left(a_v b_i - b_v a_i\right). \tag{3.3}$$

A lagging (inductive) current gives a positive Q_1. The total active power P is the mean of the instantaneous product $v_n i_n$ over the window, and $S = V_{\mathrm{rms}} I_{\mathrm{rms}}$. The remaining indices are

$$\mathrm{PF} = \frac{P}{S}, \qquad \mathrm{DPF} = \cos\varphi_1 = \frac{P_1}{\sqrt{P_1^2 + Q_1^2}}, \tag{3.4}$$

$$D = \sqrt{S^2 - P^2 - Q_1^2}, \qquad \mathrm{THD}_I = \frac{\sqrt{\sum_{h=2}^{15} I_h^2}}{I_1}. \tag{3.5}$$

For a nearly sinusoidal voltage, $\mathrm{PF} \approx \mathrm{DPF}/\sqrt{1 + \mathrm{THD}_I^2}$. This relation is used in Chapter 4 as a consistency check on the reported values.

3.5.3 Noise Separation and Deadbands

Let the true current s repeat every cycle while the sensor noise n is random. Over $K = 5$ cycles, the mean square of the raw samples is $\overline{x^2} = S^2 + N^2$. The mean square of the averaged cycle is $\overline{\bar{x}^2} = S^2 + N^2/K$. Solving these two equations gives

$$S^2 = \frac{K\,\overline{\bar{x}^2} - \overline{x^2}}{K - 1}, \qquad N^2 = \overline{x^2} - S^2. \tag{3.6}$$

The firmware reports $\sqrt{S^2}$ as the current rms and $\sqrt{N^2}$ as the sensor noise, with no separate calibration. Active power is not affected by the noise, because random noise does not correlate with the voltage.

Below fixed thresholds, a quantity and everything derived from it are forced to zero. The thresholds are 20 V rms for the voltage (treated as mains off), 0.02 A for the grid current after noise removal, and 0.05 A for sensor A, which applies to the

display only and only while the bridge is idle.

3.5.4 Digital Filters

Two second-order Butterworth low-pass filters are designed with the bilinear transform and pre-warping ($K = \tan(\pi f_c / f_s)$) [13]. They are implemented in direct form II transposed, which needs only two state variables per filter:

- **Sensor A, causal, $f_c = 1\,\mathbf{kHz}$.** The 15th harmonic at 750 Hz passes, but with gain 0.878 and a phase shift of $-66.5°$. Because harmonic h always sits at the normalised frequency $2\pi h/200$, the firmware computes the exact complex response $H(e^{j\omega_h})$ once at start-up and divides every measured AHF phasor by it. The control loop therefore receives the true phasor.
- **V and I display, zero-phase, $f_c = 1.5\,\mathbf{kHz}$.** The averaged cycle is periodic, so the filter is run forwards and then backwards around it, and the phase shifts cancel. This filter is applied only to the waveforms sent to the screen. Every number (rms, P, Q, PF, THD, harmonics) is computed from the unfiltered data first.

3.6 Automatic Power Factor Correction

Every 0.5 s, using measurements averaged over the previous five windows, the APFC controller:

1. applies the manual locks directly if all three steps are locked by the operator (hold reason 3);
2. holds if $V < 10\,\mathrm{V}$ or $|P_1| < 0.5\,\mathrm{W}$, which is treated as no load (hold reason 1);
3. otherwise computes the reactive power per microfarad and the capacitance needed:

$$k = 2\pi f V^2 \times 10^{-6}, \qquad Q_{\mathrm{load}} = Q_1 + k\,C_{\mathrm{on}}, \tag{3.7}$$

$$Q_{\mathrm{tgt}} = |P_1| \tan(\arccos 0.99), \qquad C_{\mathrm{need}} = \max\!\left(0, \frac{Q_{\mathrm{load}} - Q_{\mathrm{tgt}}}{k}\right); \tag{3.8}$$

4. evaluates all $2^3 = 8$ combinations allowed by the manual locks and chooses the one closest to C_{need}, breaking ties in favour of the fewest relay changes;
5. switches only if the present combination is off by more than $0.75\,\mu\mathrm{F}$ and the best combination is better; otherwise it holds and reports why. Hold reason 2 means that the need is less than half the smallest step.

At 230 V and 50 Hz, $\tan(\arccos 0.99) = 0.1425$. The term $k\,C_{\text{on}}$ adds back the capacitors already switched in. Without it, switching a capacitor in would lower the measured Q_1 and the controller would switch it straight out again. Every step, automatic or manual, also observes a minimum ON time and a minimum OFF time of 5 s.

3.7 Active Harmonic Filter Control

3.7.1 Why a Direct Copy of the Distortion Fails

An early approach scaled the distortion waveform $-(i - i_1)$ to the full duty range and played it on the PWM. This fails for three reasons. The bridge produces a voltage, so the current through the series inductance is that voltage divided by $jh\omega L$: the 15th harmonic comes out five times weaker than the 3rd, and every harmonic lags by nearly 90°. The amplitude is also fixed, regardless of the current required. And because nothing is measured, errors are never corrected. In simulation this method increased the THD from 27.9% to 61.5% [16].

3.7.2 From Phasors to PWM

Each harmonic $h = 2, \ldots, 15$ has a complex duty command U_h, which the controller updates once per clean 100 ms window. The duty command for one cycle is synthesised from these phasors:

$$u_k = \sum_{h=2}^{15} \operatorname{Re}\left\{U_h e^{j2\pi hk/200}\right\}, \qquad k = 0, \ldots, 199. \tag{3.9}$$

If the peak of u would exceed 0.92, all U_h and u are scaled down together. This anti-windup step keeps the integrators from growing while the output is saturated. The table is stored in Q15 format and double-buffered, and the TIM2 ISR outputs one value per sample. Because the table is synthesised only from $U_2, \ldots, U_{15}$, noise and content above the 15th harmonic are not amplified.

3.8 Communication Protocol

USART2 runs at 115 200 bit/s. With BRR = 278 at 32 MHz, USARTDIV is 17.375 and the actual rate is 115 108 bit/s, an error of −0.08%. Transmission uses DMA1 Stream 6, Channel 4. Reception uses an interrupt-driven line parser with strict range checks. All messages are ASCII lines that start with `$` (Table 3.6).

A frame is about 4 kB, and most of it is the five waveforms. At about 11 520 bytes/s,

Table 3.6: USART2 text protocol between the firmware and the dashboard

Line	Content
PC to MCU (commands)	
`$C,i,0,0`	Return capacitor step i (0, 1, 2) to automatic
`$C,i,1,s`	Lock step i manually ON ($s = 1$) or OFF ($s = 0$)
`$A,e`	AHF ACTIVE ($e = 1$) or STANDBY ($e = 0$)
MCU to PC (one frame every 0.5 s)	
`$W,V/I/A/T/C`	100 points of one cycle: voltage, grid current, AHF current, AHF target, predicted current after compensation ($I + A$)
`$M`	V_{rms}, I_{rms}, P, S, Q_1, PF, THD_V, THD_I, crest factor, peaks, DPF, D, frequency
`$H,I/A/C`	RMS of harmonics 1–15 of the grid current, the AHF current and the predicted compensated current
`$P`	Relay state, capacitance on, PF, Q_1, hold flag, target, C_{need}, PLL lock, hold reason, steps waiting for lock
`$A`	AHF active, AHF peak, target peak, duty peak, loop residual, THD after
`$K`	Manual lock mask and state, commands accepted / rejected, bytes received
`$R`	Per-channel ADC offset, minimum, maximum, clip count; overruns, dropped windows, skipped frames, current-sensor noise
`$E`	End of frame: the PC applies everything received since the last `$E` at once

one frame takes roughly 0.36 s of the 0.5 s between frames. This is why the waveforms are decimated to 100 points and why the firmware counts a skipped frame if the previous DMA transfer has not finished. A text protocol is slower than a binary one, but it can be read in any serial terminal, and it resynchronises at every newline.

3.9 PC Dashboard

The dashboard is a Python program built on PySide6 (Qt 6), pyqtgraph, numpy and pyserial. A background reader thread reads the serial port, splits the byte stream into lines and passes them to a telemetry store. The store collects the lines of one frame and commits them under a lock only when `$E` arrives, so the display never mixes the waveforms of one frame with the numbers of the next. The GUI thread refreshes at about 30 frames per second from a snapshot of the store. The window has two tabs:

- **Power Factor Correction:** PF gauge with the DPF marker and the 0.99 target; P, Q_1, S, D, V, I, capacitance on and capacitance required; the capacitor-bank card with automatic and manual control and the hold explanation; PF and capacitance trends.
- **Harmonic Mitigation:** THD before and after, THD_V, crest factor, AHF tracking correlation, spectrum shape match, H1 leak and I_1; the AHF card with ACTIVE/STANDBY buttons and diagnostics; AHF current against target; grid current before and after; grid and AHF harmonic spectra.

Every electrical quantity is computed on the microcontroller. The PC computes only display aids: the correlation between the AHF current and its target, the spectrum shape match, the H1 leak and the percentage reduction. A demo mode generates the same line protocol without hardware, which exercises the parser and the GUI.

3.10 Hardware and Software Requirements

Table 3.7 lists the hardware and software used in the prototype.

Table 3.7: Hardware and software requirements

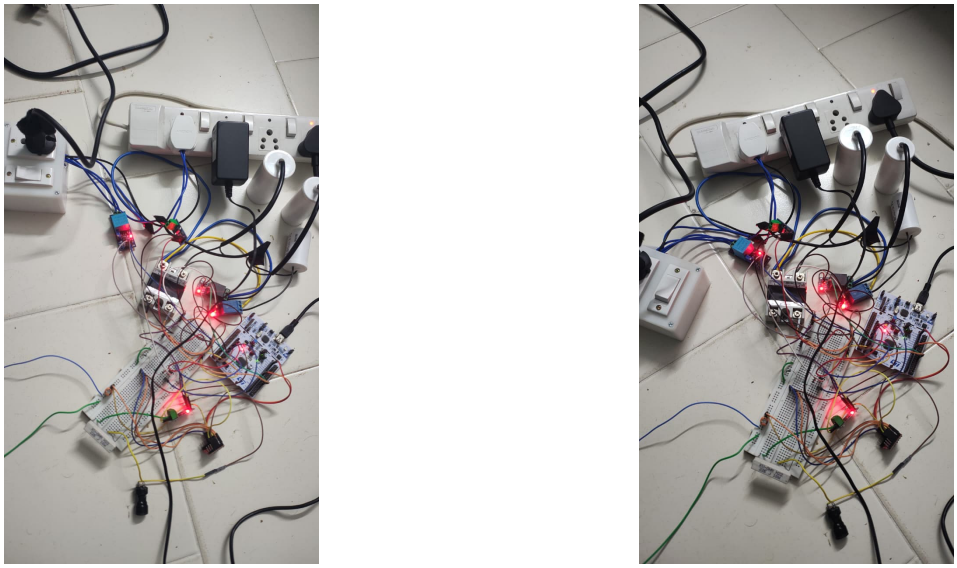
Item	**Role**
NUCLEO-F401RE (STM32F401RE)	Measurement, control and communication
ZMPT101B module	Mains voltage sensing
2 × ACS712-05B modules	Grid current and AHF current sensing
10 kΩ / 20 kΩ divider	Level shifting of the grid-current sensor output
Capacitors 2.5, 10, 20 μF (400 VAC)	Switched reactive compensation
2 relay modules (5 V, active-LOW, 10 A), 1 zero-cross SSR	Capacitor switching
TB6612FNG H-bridge (VCC 3.3 V, VM 12 V)	AHF output stage
10 mH inductor, 10 W resistor	AHF test circuit, not connected to the mains (resistance not recorded)
C with CMSIS headers, GCC (-O2)	Bare-metal firmware (no HAL, no RTOS)
Python with PySide6, pyqtgraph, numpy, pyserial	Dashboard and demo simulator

CHAPTER 4

Results and Discussions

4.1 Experimental Setup

Fig. 4.1 shows the bench prototype. Visible are the NUCLEO-F401RE board connected to the PC by USB, the sensor and relay modules, the capacitor bank, a breadboard carrying the low-voltage wiring, and the mains distribution boards used to connect the load under test. Fig. 4.2 shows the same setup during a test with a hair dryer as the load, with the dashboard running on the laptop.



(a) View 1 **(b)** View 2

Figure 4.1: Hardware experimental setup: NUCLEO-F401RE board, sensor and relay modules, capacitor bank and breadboard wiring

The measurements in this chapter were taken on the bench prototype from the single-phase mains. For each snapshot the dashboard reports the supply voltage, the frequency and the lock state; these are quoted with the results. The loads were a laptop adapter, a load drawing 0.69 A, a load drawing 0.204 A, a phone charger and a hair dryer at normal speed. The laptop adapter was tested in two separate sessions. Instrument cross-checks against a reference power analyser are not documented, and calibration is nominal or single-point (Section 4.7.1).

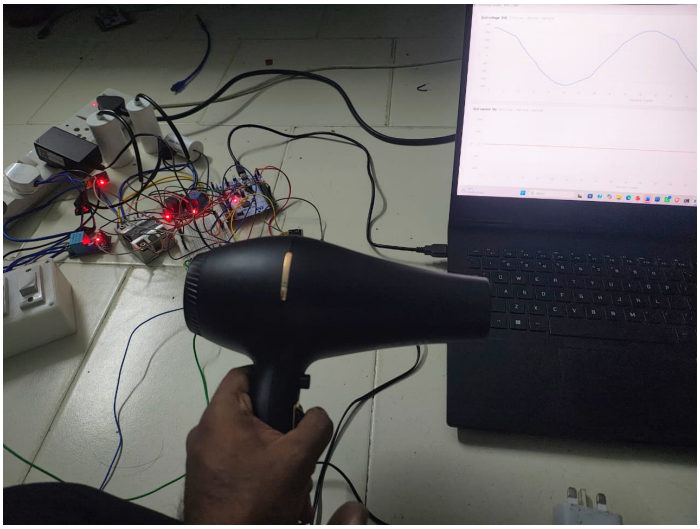


Figure 4.2: Test setup with a hair dryer as the load under test and the dashboard running on the laptop

4.2 Basis of Evaluation

All results come from a small number of bench tests on single-phase mains. They show how the prototype behaved on the particular loads tested, not a general performance figure. Three points apply throughout:

- **THD before compensation** is measured by sensor I.
- **THD after compensation** is a *prediction*. The AHF is not connected to the mains (Section 3.3.4). The MCU computes the per-harmonic sum of the measured load current and the measured AHF output current, $I_h + A_h$. Both terms are real measurements, but their sum is the grid current that *would* flow if this AHF current were injected. No compensated current flows in the prototype.
- **What is directly demonstrated** is that the bridge produces the required opposite harmonic current. This is shown by the tracking correlation between the measured AHF current and its target, by the spectrum shape match and by the loop residual.

Values read from the dashboard bar charts are approximate and are marked “about”.

4.3 Summary of Harmonic Results

Table 4.1 collects the THD results. The first four rows are the results recorded in the study guide [16]. The last two rows are from the dashboard snapshots in Figs. 4.5 and 4.9.

Table 4.1: Current THD before and predicted after AHF compensation (bench tests)

Load	THD_I before	THD_I after (predicted)	Note
Laptop adapter (session 1)	35.9%	10.5%	PF about 0.9 without capacitors; small Q_1
0.69 A load	15.9%	3.2%	–
0.204 A load	33.7%	14.4%	Low current, close to sensor resolution
Phone charger	–	–	At the noise floor of the ACS712
Laptop adapter (session 2)	42.7%	5.8%	Fig. 4.5; 41.9% → 6.4% in a later frame
Hair dryer, normal speed	53.5%	8.1%	Fig. 4.9; 53.1% → 10.9% in a later frame

In the first three tests the H-bridge used only 7–12% of its duty range [16]. In the two later sessions the dashboard reports a duty peak of 28–29%. The THD values change from frame to frame, as the paired frames in Table 4.1 show. A single frame should therefore not be read as a settled value.

4.4 Laptop Adapter (Session 2)

4.4.1 Waveforms and Power Quantities

Fig. 4.3 shows the voltage and current over two cycles. The dashboard reports 252.2 V rms at 49.13 Hz with a voltage THD of 3.6%. The current is 0.672 A rms with a THD of 39.5% and a crest factor of 1.97. The current has the stepped, pulsed shape of a rectifier front end and flattens near zero between the conduction intervals.

The Power Factor Correction tab (Fig. 4.4) shows a later frame: $P = 132.6\,\mathrm{W}$, $Q_1 = +31.6\,\mathrm{var}$ (lagging), $S = 149.1\,\mathrm{VA}$ and $D = 60.5\,\mathrm{VA}$, with PF 0.889 and DPF 0.973 at 252.4 V and 0.591 A. Two consistency checks, computed by the authors from the displayed values, agree with the firmware's outputs:

- $\sqrt{S^2 - P^2 - Q_1^2} = \sqrt{149.1^2 - 132.6^2 - 31.6^2} \approx 60.4\,\mathrm{VA}$, which matches the reported D;
- $\mathrm{DPF}/\sqrt{1 + \mathrm{THD}_I^2} = 0.973/\sqrt{1 + 0.427^2} \approx 0.895$, close to the reported PF of



Figure 4.3: Laptop adapter: grid voltage and grid current over two cycles (dashboard)

0.889.

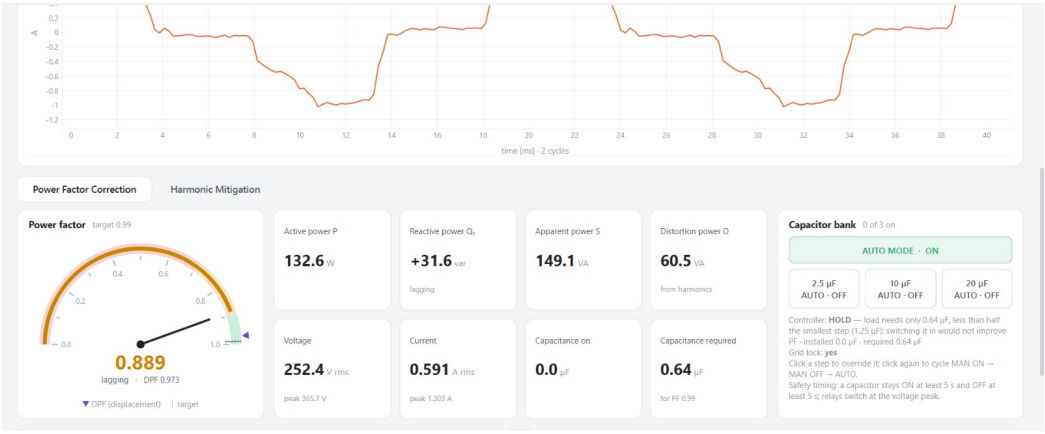


Figure 4.4: Laptop adapter: Power Factor Correction tab showing the controller holding with no capacitor switched in

4.4.2 APFC Behaviour

No capacitor is on, and the controller reports HOLD because the load needs only $0.64\,\mu$F, less than half of the smallest step. The value follows from the algorithm. At $252.4\,$V, $k \approx 20.0\,\text{var}/\mu\text{F}$ and $Q_{\text{tgt}} = 132.6 \times 0.1425 \approx 18.9\,\text{var}$, so $C_{\text{need}} = (31.6 - 18.9)/20.0 \approx 0.64\,\mu$F. Switching in the $2.5\,\mu$F step would add about $50\,$var of leading reactive power and overshoot the target. Here the low PF comes mainly from distortion ($D = 60.5\,$VA), not from displacement.

4.4.3 AHF Behaviour

In the Harmonic Mitigation tab (Fig. 4.5), the current THD falls from 42.7% before compensation to a predicted 5.8% after. The measured AHF current tracks its target with a correlation of $\rho = 0.976$ (peak 0.424 A against a target peak of 0.482 A). The spectrum shape match is 100%, the H1 leak (fundamental in the AHF current) is 6%, and the fundamental is $I_1 = 0.510$ A. The diagnostics block shows no ADC clipping, no overruns, no dropped windows and no skipped frames, and the loop residual is 12.7%.

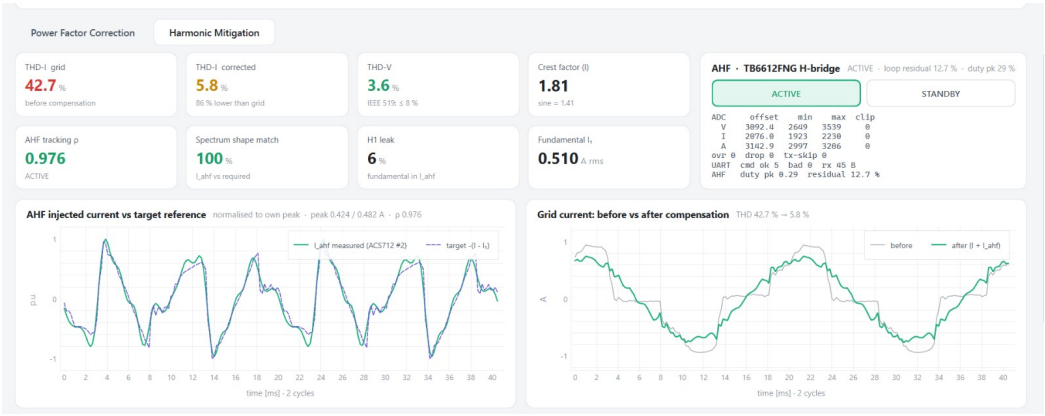


Figure 4.5: Laptop adapter: Harmonic Mitigation tab with AHF current against target and grid current before and after ("after" is the prediction $I + A$)

Fig. 4.6 shows the spectra from a later frame (41.9% → 6.4%). Before compensation, the grid current is dominated by odd harmonics: the 3rd is at about 37% of I_1, the 5th about 15%, the 9th about 10% and the 7th and 13th about 5–6%. In the predicted after spectrum every harmonic is about 4% or less. The measured AHF spectrum closely follows the required harmonic pattern, with the 3rd harmonic largest.

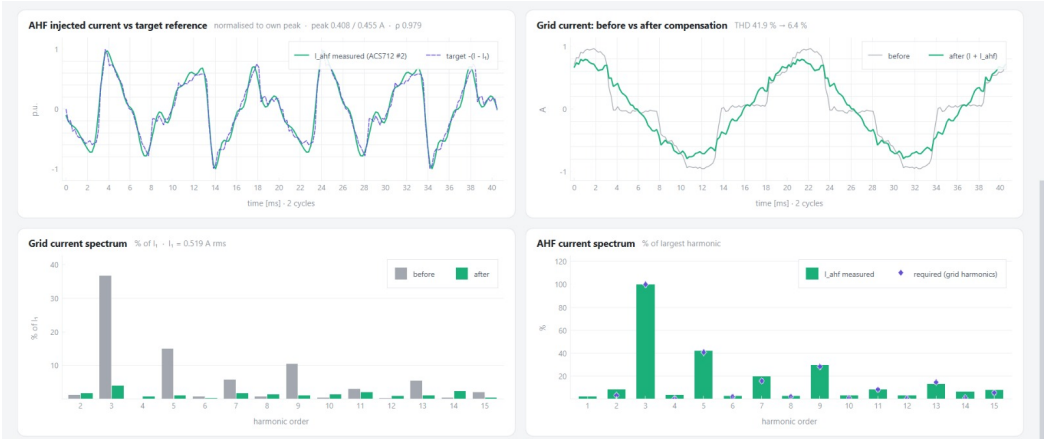


Figure 4.6: Laptop adapter: grid-current spectrum before and after (predicted) and AHF current spectrum against the required harmonics

4.5 Hair Dryer at Normal Speed

4.5.1 Waveforms and Power Quantities

Fig. 4.7 shows 252.6 V rms at 49.12 Hz with a voltage THD of 3.5%. The current is 0.464 A rms with a THD of 53.4% and a crest factor of 2.15. The current waveform is clearly asymmetric: its positive peak is about 0.6 A and its negative peak about −0.9 A. This is consistent with the large second harmonic seen in the spectrum below. The cause of the asymmetry inside the appliance was not investigated.



Figure 4.7: Hair dryer (normal speed): grid voltage and grid current over two cycles (dashboard)

4.5.2 APFC Behaviour

In Fig. 4.8 the controller has switched the 2.5 μF step in automatically. With it in circuit, the dashboard shows $P = 99.7$ W, $Q_1 = +21.9$ var, $S = 116.4$ VA and $D = 55.9$ VA, with PF 0.857 and DPF 0.977. The required capacitance is 2.89 μF. This follows from the algorithm: at 253.1 V, $k \approx 20.1$ var/μF, $Q_{\text{load}} = 21.9 + 2.5k \approx 72.2$ var and $Q_{\text{tgt}} \approx 14.2$ var, so $C_{\text{need}} \approx 2.89\,\mu$F. The closest combination is the 2.5 μF step already on. The remaining error of 0.39 μF is inside the 0.75 μF deadband, so the controller holds. The consistency checks give $\sqrt{116.4^2 - 99.7^2 - 21.9^2} \approx 55.9$ VA and $0.977/\sqrt{1 + 0.535^2} \approx 0.861$, against a reported PF of 0.857.

4.5.3 AHF Behaviour

Fig. 4.9 shows the current THD falling from 53.5% to a predicted 8.1%. The tracking correlation is $\rho = 0.982$ (AHF peak 0.435 A against a target peak of 0.432 A), the H1

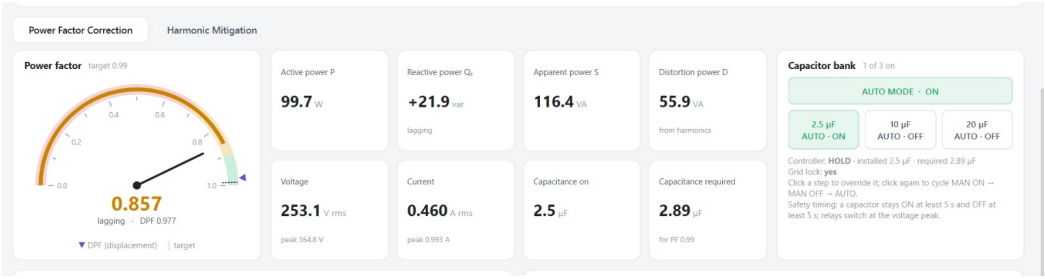


Figure 4.8: Hair dryer: Power Factor Correction tab with the 2.5 μF step on and the controller holding

leak is 8%, the duty peak is 28% and the loop residual is 15.0%, with no clipping, overruns or dropped windows.

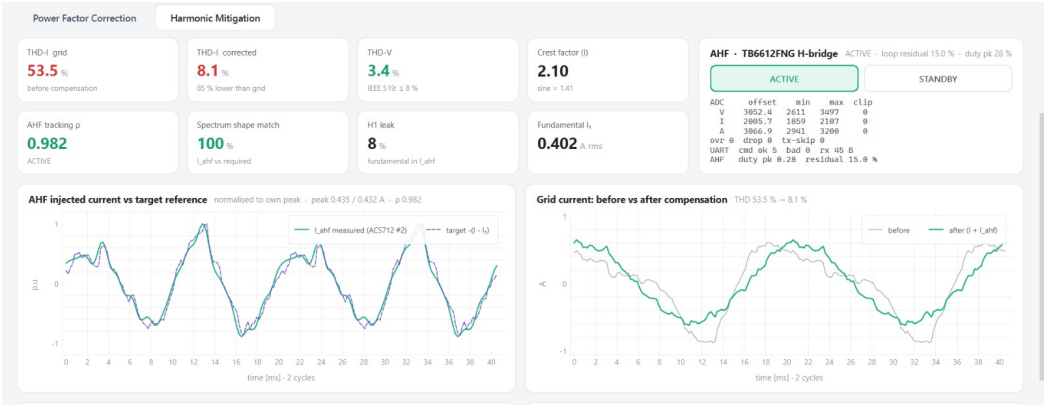


Figure 4.9: Hair dryer: Harmonic Mitigation tab ("after" is the prediction $I + A$)

The spectra in Fig. 4.10 come from a later frame (53.1% → 10.9%). Before compensation, the second harmonic is about 50% of I_1 and the third about 15%. The predicted after spectrum removes most of the 2nd and 3rd harmonics. The 5th harmonic, however, is *higher* after compensation (about 9%) than before (about 1–2%). In the AHF spectrum the measured 5th-harmonic current is about 20% of the largest harmonic, while the required value is only a few per cent. In this frame the 5th-harmonic loop therefore produced more than was needed. The available data do not establish the cause. Possible contributors include the controller's plant-gain estimate at the 5th harmonic and the interaction with the strong 2nd harmonic, but this was not investigated further.

4.6 APFC Behaviour on Other Loads

4.6.1 Laptop Adapter (Session 1)

The study guide records a dashboard reading of $V = 246\,\mathrm{V}$, $P_1 = 97.5\,\mathrm{W}$ and $Q_1 = 15.3\,\mathrm{var}$ with no capacitor on [16]. Here $k = 19.0\,\mathrm{var}/\mu\mathrm{F}$ and $Q_{\mathrm{tgt}} = 13.9\,\mathrm{var}$, so

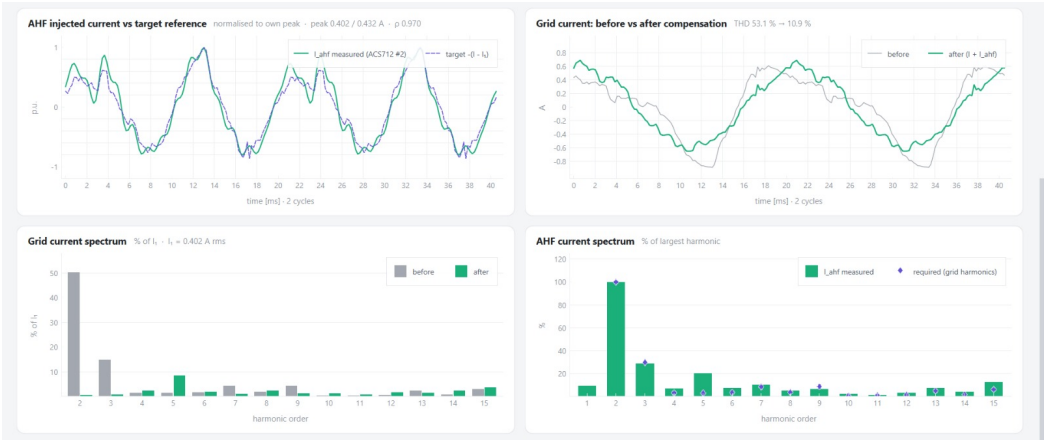


Figure 4.10: Hair dryer: grid-current spectrum before and after (predicted) and AHF current spectrum against the required harmonics

$C_{\text{need}} = 0.07\,\mu\text{F}$. The controller held with reason 2. Switching in the $2.5\,\mu\text{F}$ step would have added about 47.5 var of leading reactive power and made the PF worse. In another snapshot of this adapter, the guide records $S = 105.7\,\text{VA}$, $D = 37.7\,\text{VA}$, DPF 0.988, THD 34% and PF 0.923. This again shows that the penalty was almost entirely due to distortion.

4.6.2 Phone Charger

The phone charger showed a PF of about 0.5 but a Q_1 of only about 3 var. This corresponds to a need of about $0.2\,\mu\text{F}$, far below the smallest step. When the $2.5\,\mu\text{F}$ step was switched in manually, Q_1 became about -38 var (leading) [16]. The low PF of this load is therefore caused by distortion, not displacement, and a capacitor makes it worse. The automatic controller correctly holds in this case.

4.7 Discussion

The results answer the research questions of Chapter 1 for the loads tested. The measurement chain separated displacement from distortion clearly. The laptop adapter had a DPF of 0.973–0.988 but a PF of only 0.889–0.923, and the phone charger had a PF of about 0.5 with almost no reactive power. For these electronic loads the power factor penalty is mostly distortion, which a capacitor bank cannot correct. This supports the two-stage design.

The capacitor controller made the expected decision in every documented case. It switched the $2.5\,\mu\text{F}$ step in for the hair dryer, whose demand justified it, and it held for both laptop sessions and the phone charger, where a step would have overcompensated. Its reported hold reasons matched values recomputed by hand from the displayed quantities.

The AHF loop produced the required opposite harmonic current in its test circuit. The measured current tracked the target with a correlation of 0.976–0.982 and a spectrum shape match of 100%, while using only 28–29% of the duty range. The predicted THD reductions are therefore limited mainly by the loop residual (12.7–15.0%) rather than by the output stage. The hair dryer's 5th harmonic shows that one per-harmonic loop can overshoot while the others converge, and this needs further investigation.

4.7.1 Limitations

- **No grid injection.** The AHF drives a separate test circuit and is not connected to the mains. The after-compensation THD is a prediction calculated as $I + A$. It assumes the measured AHF current would add to the grid current unchanged if injected, which was not tested. There is no grid-side sensor, and the polarity `A_SIGN` has not been confirmed against the grid current.
- **AHF power stage.** The TB6612FNG is a motor driver rated 1.2 A continuous (3.2 A peak) [17], supplied at 12 V. It demonstrates the control method but is not a grid-rated inverter. A grid-connected filter would need an inverter with a DC bus above the grid peak and a proper isolation and coupling design.
- **Harmonic range.** Only the 2nd to the 15th harmonics are measured and compensated.
- **Sensor resolution.** The grid-current resolution is 6.5 mA per count, so small loads such as the phone charger sit near the noise floor.
- **Capacitor step size.** The smallest step is 2.5 μF, about 41 var at 230 V, so small reactive demands cannot be corrected without overshoot.
- **Relay inrush.** The two mechanical relays have no NTC inrush limiting. Only the 20 μF step uses a zero-cross SSR.
- **Single phase only.** Calibration of K_V and of the ACS712 sensitivities is nominal or single-point.
- **Number of tests.** Only a few loads and sessions were tested. The results cannot be generalised to other loads or operating conditions.

CHAPTER 5

Conclusion & Future Scope

5.1 Conclusion

This project built a dual-stage power quality system on an STM32F401RE using register-level, bare-metal firmware. A timer-triggered ADC with DMA double buffering samples one voltage and two currents at exactly 200 samples per cycle. A software PLL keeps the sampling locked to the grid, so that each 100 ms window holds five whole cycles. On this basis the firmware computes the power quantities that distinguish displacement from distortion and drives two actuators. The first is a capacitor bank, with a decision rule that knows when to hold, together with minimum on/off times and point-on-wave switching. The second is a selective-harmonic AHF loop that identifies its own plant gain at each harmonic and drives an H-bridge with 80 kHz sign-magnitude PWM.

On the loads tested, the capacitor controller behaved as designed. It switched a step in when a lagging load justified it (the hair dryer) and held when the reactive demand was below half of the smallest step (the laptop adapters and the phone charger). The AHF loop made the bridge produce the required opposite harmonic current in its test circuit, with a tracking correlation of 0.976–0.982 in the dashboard snapshots. When this measured current is added to the load current, the predicted current THD falls in every test where the load current was above the sensor noise floor. Examples are 42.7% to 5.8% for the laptop adapter and 53.5% to 8.1% for the hair dryer. These after-compensation values are computed predictions ($I + A$), because the AHF is not connected to the mains. The tests are few, and the H-bridge is a low-power demonstration stage. The work therefore demonstrates the embedded measurement and control method, not grid-ready performance.

5.2 Future Scope

The most direct next step is to connect the AHF to the mains through a grid-rated inverter and coupling stage. A grid-side current sensor would then measure the compensated current directly instead of predicting it, and would confirm `A_SIGN`. The study guide identifies the following extensions [16]:

- a grid-side current sensor to measure the compensated current directly;

- an IGBT or MOSFET inverter with a 400 V DC bus in place of the TB6612FNG;
- compensation of more harmonics, using a faster ADC;
- thyristor-switched capacitors in place of mechanical relays;
- extension to three-phase systems;
- data logging; and
- measurement compliant with IEC 61000-4-7 [8].

Before further capacitor-switching experiments, NTC inrush limiters should be fitted.

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Appendix 1 – Mapping the Project with POs, PSOs and SDGs

Title of the Project: Automatic Power Factor Correction & Harmonic Mitigation System
Course: 23ELC302 – Embedded System Design using ARM

Course Outcomes

Table A.1: Mapping of the project with the Course Outcomes

CO	Course Outcome	Level	Justification to the Contribution
CO1	Understand the basics of embedded systems.	2	The project applies the basic elements of an embedded system: sensors, signal conditioning, an MCU, actuators (relays, SSR, H-bridge) and a host interface, working together under real-time constraints.
CO2	Comprehend embedded computing architecture.	3	The firmware relies on the Cortex-M4F architecture: the FPU, the AHB/APB bus and clock tree, the NVIC priority scheme, DMA streams and timer trigger routing (Tables 3.4 and 3.5).
CO3	Develop programs for ARM based microcontrollers.	3	Register-level CMSIS firmware without HAL configures the clocks, GPIO, TIM2, TIM4, ADC1, DMA and USART2, and implements the DSP and both control loops (Listings 3.1 and 3.2).
CO4	Demonstrate ARM based real time applications through simulation and hardware.	3	The system runs in real time on the bench with mains loads (Chapter 4). Simulation studies of the PLL, the AHF loop and relay switching supported the design.

Correlation level: 3 = substantial, 2 = moderate, 1 = slight.

PO Mapping – Justification for this Project

Table A.2: Mapping of the project with Program Outcomes

PO	Outcome	Level	Justification to the Contribution
PO1	**Engineering Knowledge:** Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.	3	Mathematics (DFT, phasors, complex plant gains, bilinear transform), power engineering fundamentals (P, Q_1, S, D, PF, DPF) and computing (register-level firmware) are combined to separate displacement from distortion and to design both correction stages (Sections 3.5 and 3.3.4).
PO2	**Problem Analysis:** Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4)	3	The problem was formulated from the behaviour of real loads. For example, the low PF of the phone charger and laptop adapter comes from distortion, not displacement, so capacitors cannot correct it. Power quality definitions, harmonic standards and active filter literature were reviewed (Chapter 2). Conclusions are backed by consistency checks of the reported quantities (Chapter 4).
PO3	**Design/Development of Solutions:** Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5)	3	A two-stage system was designed and built. It has capacitor selection with hysteresis and hold reasons, and a selective-harmonic AHF loop with online plant identification. Safety is considered through minimum on/off times and point-on-wave switching that protect the relay contacts, and through keeping the low-voltage AHF stage off the mains. The design uses low-cost, readily available modules (Chapter 3).

Table A.2 continued

PO	Outcome	Level	Justification to the Contribution
PO4	**Conduct Investigations of Complex Problems:** Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8).	2	Bench tests were run on several loads across two dashboard sessions. Simulation studies covered the PLL, AHF convergence and relay switching. Faults were diagnosed: sensor clipping led to the divider, and relay welding led to the switching rules. The data are interpreted with their limits stated, including measured versus predicted values and sensor resolution (Chapters 3 and 4).
PO5	**Engineering Tool Usage:** Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6)	3	The firmware is register-level CMSIS C compiled with GCC. The Python dashboard uses PySide6, pyqtgraph, numpy and pyserial and includes a demo simulator. The design applies model-based prediction (the initial plant model, online identification, and the $I + A$ prediction), and the report states the limits of that prediction (Sections 3.3.4 and 4.7.1).
PO6	**The Engineer and The World:** Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7).	1	The project considers how power quality affects line current and losses (Table A.4), and the safety of switching capacitors on the mains: relay inrush and welding, the zero-cross SSR, and the recommended NTC limiters. The relevant standards are cited without claiming compliance (Chapter 2).
PO7	**Ethics:** Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9)	1	Results are reported honestly. Measured and predicted values are kept separate, findings from a few tests are not generalised, no compliance or SDG-target claims are made, unexpected results such as the hair dryer's 5th harmonic are disclosed, and all borrowed material is cited.

Table A.2 continued

PO	Outcome	Level	Justification to the Contribution
PO8	**Individual and Collaborative Team work:** Function effectively as an individual, and as a member or leader in diverse/multidisciplinary teams.	2	The project was carried out by a three-member team. It joins work from several disciplines: mains-side hardware, embedded firmware, signal processing and PC software.
PO9	**Communication:** Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences	2	The work is documented in this report and in a detailed study guide prepared for the project presentation and panel review. The dashboard presents the results, and the reason for every capacitor hold, in plain language.
PO10	**Project Management and Finance:** Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.	1	Economic choices are visible in the design: a low-cost NUCLEO board, sensor modules, and a motor-driver H-bridge as a demonstration stage in place of a grid-rated inverter. The work was developed in stages and the firmware revised iteratively.
PO11	**Life-Long Learning:** Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change. (WK8)	2	The team learned register-level STM32 programming without HAL, as well as DSP and closed-loop control. It adapted the design when methods failed: the direct-copy AHF method, positive-current clipping and relay welding each led to a revised approach. Future extensions are identified (Chapter 5).

PSO Mapping

Table A.3: Mapping of the project with Program Specific Outcomes

PSO	Outcome	Level	Justification to the Contribution
PSO1	Design and develop autonomous systems by integrating knowledge of electrical and computer engineering domains.	3	The STM32 measures, decides and acts on its own. It keeps correcting with the PC disconnected. It integrates electrical engineering (power theory, capacitor compensation, H-bridge drive) with computer engineering (timers, ADC/DMA, interrupt priorities, bare-metal firmware) (Chapter 3).
PSO2	Apply computational and networking skills for secured smart electrical systems.	1	Computational skills: real-time DSP on the MCU (folding, DFT, noise separation, filters, PLL). Communication: a USB/UART link with framed telemetry, atomic frame commit, and strict range-checked commands with accepted/rejected counters. On security, only input validation is implemented; the link has no authentication or encryption (Table 3.6).
PSO3	Innovate sustainable solutions for interdisciplinary domains using emerging technologies and data-driven approaches.	2	The solution brings together power engineering, embedded systems, signal processing, control and software. It is data-driven: capacitor decisions come from measured, averaged quantities, and the AHF identifies its plant gain at each harmonic from measured data. Supplying reactive power locally supports more efficient use of electricity (Table A.4).

SDG Mapping

Table A.4: Mapping of the project with Sustainable Development Goals

SDG	Goal	Justification to the Contribution
SDG 7	**Affordable and Clean Energy** *Ensure access to affordable, reliable, sustainable and modern energy for all by 2030.*	The APFC stage supplies the reactive power of a lagging load locally, so the supply carries less current for the same active power. Lower line current means lower resistive losses in the wiring [2]. The controller also avoids overcompensation: it holds when the need is less than half the smallest step. The AHF work shows how the extra current caused by harmonics could be reduced, using low-cost hardware. The contribution is modest and was shown only on a bench prototype; the energy saved was not measured.
SDG 9	**Industry, Innovation and Infrastructure** *Build resilient infrastructure, promote inclusive and sustainable industrialization and foster innovation by 2030.*	The project develops and documents a low-cost embedded approach to power quality. Grid-synchronised measurement, capacitor switching that protects the contacts, and selective harmonic control all run on one microcontroller. Better power quality reduces stress on wiring and equipment, which supports reliable electrical infrastructure. This is a prototype-level innovation, not a deployed infrastructure upgrade.

The project is a single-phase bench prototype and does not claim to achieve any SDG target.